Living Earth Cube-in-a-Box - User Guide

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1 Overview

Welcome to the **Living Earth Cube-in-a-Box User Guide**. This manual is designed to help you navigate and use the Open Data Cube (ODC) platform for geospatial analysis and remote sensing.

All the developments have made possible thanks to the financial support of the European Union 'Horizon Europe Program' that funded the LandShift[1] (Grant Agreement no. 101182007), Nostradamus[2] (Grant Agreement no. 101134888), and NEMESIS[3] (Grant Agreement no. 101219087) projects.

1.1 What is Living Earth Cube-in-a-box (CiaB)?

The Living Earth CiaB is a specialized instance of the Open Data Cube[4], packaged for rapid deployment and accessibility. It provides a pre-configured environment with all the necessary tools to:

- Browse massive collections of satellite imagery.
- Run Python-based geospatial analysis using Jupyter Notebooks.
- Share results and collaborate with other researchers.

1.2 Core Platforms

To interact with the system, you will primarily use two web interfaces:

1. **Jupyter**: `http://<DOMAIN>/jupyter/`
    - Your primary analytical workspace.
    - Write and execute Python notebooks.
    - Access a persistent Linux home directory.
2. **Explorer**: `http://<DOMAIN>/explorer`
    - A visual web interface to browse available datasets.
    - Filter products by area and time.
    - View spatial and temporal coverage of indexed data.
    - Download data.

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[1] https://landshift.eu/
[2] https://nostradamus-project.eu
[3] https://www.nemesis-soil.eu/
[4] https://opendatacube.org/

1.3 How to use this guide

If you are a new user, start with the **Account Setup** section to gain access. Once authorized, explore the **Workspace** and **Explorer** sections to understand how to interact with the data.

For ready-to-use analysis templates, refer to the **Data Products** chapter which links directly to demonstration notebooks.

2 Account Setup

Access to the Living Earth CiaB is restricted to authorized users. Follow these steps to create your account and gain access.

> ⚠ Note
>
> Before you can sign up, your username must be pre-authorized by a system administrator. If you have not yet been authorized, please contact your project coordinator.

2.1 Self-Service Signup

Once you have been notified that your username is authorized:

1. Visit the **Signup Page**: `http://<DOMAIN>/jupyter/hub/signup`.
2. Enter your **Username** (must match exactly the one authorized).
3. Choose a secure **Password** (you will have to enter it twice).
4. Submit the form.

If successful, you will see a message: *"The signup was successful! You can now go to the home page and log in to the system."*

2.2 Login

1. Navigate to the **Login Page**: `http://<DOMAIN>/jupyter/hub/login`.
2. Log in with your credentials.
3. Your Jupyter server will begin to spawn. This may take a few seconds on the first run.

3 Your Jupyter Workspace

Once logged in, you are presented with the **JupyterLab**[1] interface. A powerful, web-based environment for interactive computing and geospatial analysis.

> **Note**
>
> Non-admin users are allowed to perform `git` or `conda` operations. However, they are not allowed to perform any operation that modifies the system-wide environment (such as `sudo apt install`).
> In case of real need `admin`right can be exceptionally granted, under request to the system administrator).

3.1 Interface Basics

- **File Browser** (Left Sidebar): Manage your files and directories.
- **Main Work Area**: Where your notebooks, terminals, and text files appear.
- **Top Menu**: Access file operations, kernel management, and settings.

3.2 Data Persistence

All files stored in your JupyterLab root directory (`/notebooks/`, or `./` from the perspective of the JupyterLab interface) are **persistent**. This means your work, custom scripts, and installed packages will remain available across sessions.

3.3 Data Backup & Restore

While the system volumes are robust, it is a best practice with remote storage to periodically back up your important notebooks and results.

Option 1: Manual Backup (Download)

1. Right-click on a file or directory in the sidebar.
2. Select **Download** or **Download Current Folder as an Archive** (respectively).
3. Save the file to your local machine.

Option 2: Terminal Backup (Advanced) You can create a compressed archive of your workspace using the terminal:

```
tar -czvf my_backup_2026.tar.gz /any_directory/
```

[1] https://jupyterlab.readthedocs.io/en/stable/getting_started/overview.html

Then download the resulting `.tar.gz` file.

Git

Users familiar with `git`[2] can use it to back up their work to a remote repository.

3.4 Managing Your Server

If your environment feels sluggish:

1. Go to **File** -> **Hub Control Panel**.
2. Click **Stop My Server**.
3. This frees up system memory and CPU for other users while keeping your data safe.

! Note

When you have finished your work for the day: **File** -> **Log Out**. To preserve resources for other users.

[2] https://github.com/git-guides

4 Exploring Data

The **CiaB Explorer** is a visual web interface (`http://<DOMAIN>/explorer`) that allows to discover what data is currently indexed and available without writing a single line of code.

4.1 Key Features

- **Product Metadata**: View detailed descriptions of each dataset (e.g., Sentinel-2, Landsat).
- **Spatial Extent**: See a map overview of the geographic areas covered by the data.
- **Temporal Coverage**: Check the available time ranges for each product.
- **Individual Scenes**: Click through to see individual granules and their specific capture times.
- **Download Data**: Download data.

4.2 How to browse and optionally download data

1. **Select a Product**: Use the dropdown menu or the sidebar to choose a dataset (e.g., `s2_l2a_pc` for Sentinel-2 from Planetary Computer).
2. **Filter by Time**: Use the timeline or calendar overview to find specific dates.
3. **Inspect Metadata**: Check the `DATASETS` option, and click on any geographic box to see the scene metadata.
4. **Download Data**: Click on the urls of the files (single-band GeoTIFFs) to download in the `Location` section.

Planetary Computer assets use temporary SAS-signed HTTPS links. Copernicus Data Space (CDSE) assets are streamed through Explorer's local download proxy (short-lived signed links generated automatically). CDSE downloads require the administrator to have set `CDSE_S3_ACCESS_KEY` / `CDSE_S3_SECRET_KEY` in `.env`.

> **Tip**
>
> Use the Explorer to verify if the data you need is available before starting a heavy ODC loading process in your notebook.

5 Data Products

The Living Earth Cube comes pre-loaded with several high-value geospatial datasets. To help you get started quickly, we provide a collection of **Demonstration Notebooks** located in `./shared/notebooks_demo/`.

5.1 Available Datasets

Common products indexed in the system include:

- **Sentinel-2**: Level 2A (Surface Reflectance).
- **Sentinel-1**: RTC (Radar Backscatter).
- **Landsat**: Collection 2 Level-2.
- **Land Cover**: ESRI and ESA Worldcover products.
- **Elevation**: NASADEM.

> **Note**
>
> You will find more information in `./shared/notebooks_demo/README.md`.

5.2 Learning by Example

The best way to learn how to load and process these products is to explore the `./shared/notebooks_demo` directory in your Jupyter workspace.

5.2.1 Core Demo Notebooks:

- `Sentinel_2.ipynb`: Instructions for loading and visualizing Sentinel-2 data.
- `Sentinel_1_rtc.ipynb`: Working with Radar data.
- `Landsat_Collection_2_Level-2_Science_Products.ipynb`: Comprehensive guide for Landsat.
- `ESRI_Land_Cover.ipynb` & `ESA_Worldcover.ipynb`: Using classification maps.
- `NASADEM.ipynb`: Working with digital elevation models.

5.2.2 How to use these Notebooks:

1. Navigate to `./shared/notebooks_demo/` in your Jupyter file browser.
2. **Copy** the notebook you are interested in (Right-click -> Copy).
3. **Paste** it into jupyter root directory.
4. Open your copy and start experimenting!

Notes

- You can edit and execute `.ipynb` files directly in the `./shared/notebooks_demo` directory, but not save any changes.
- To save any changes, copy the notebook and any potential dependent files to your root directory first (or shared personal directory).

6 Collaboration

Geospatial science is often a collaborative effort. The Living Earth Cube provides specialized shared directories to facilitate teamwork and resource sharing.

6.1 The `./shared/` Directory

This directory is accessible to all users on the platform.

6.1.1 Read-Only Demonstrations Notebooks

The root of the `./shared/` directory usually contains files users want to share, and documentation (like the `notebooks_demo` directory). These files are **read-only** to ensure the core examples remain unchanged for everyone.

6.1.2 User Sharing (`./shared/all_users/`)

Under `./shared/all_users/`, you will find directories for each users.

- **Transparency**: You can browse directories and files made available by your colleagues.
- **Sharing**: To share a notebook or other files, place a copy in your own shared directory under `./shared/all_users/<your-username>/`.
- **Read-Only Access**: You can read and copy files from other users' directories, but you cannot edit or delete them.

> Best Practices for Sharing
>
> - **Communicate**: If you share a draft, add "DRAFT", a date (e.g. 20260320) or a version number to the filename.
> - **Privacy**: Only move files to `./shared/all_users/<your-username>/` that you are comfortable having the entire team see.
> - **Security**: Never share notebooks containing hard-coded API keys, passwords, or personal credentials.

7 Troubleshooting

While the system is designed to be stable, you may occasionally encounter issues. Below are solutions to common problems.

7.1 Common Issues

7.1.1 Server Won't Spawn

Symptoms: After logging in, the loading bar gets stuck, or you see a "Spawn failed" error.
Cause: The system might be under heavy load, or a previous session didn't shut down correctly.
Solution: Wait for 30 seconds and refresh the page. If it persists, go to the **Hub Control Panel** (if accessible) and click **Stop My Server**, then **Start My Server**.

7.1.2 Database Timeout ("Gateway Timeout")

Symptoms: JupyterLab loads, but running a cell with `dc.load()` or `dc.list_products()` hangs for a long time and then fails.
Cause: Usually occurs if the database is busy indexing new data or undergoing maintenance.
Solution: Restart your kernel (**Kernel** -> **Restart Kernel...**) and try again.

7.1.3 "Read-only file system"

Symptoms: You cannot save changes to a notebook.
Cause: You are likely trying to edit a file directly in the `./shared/` directory.
Solution: Copy the file to Jupyter home directory (`./`) as explained in the Collaboration chapter.

7.2 How to Reset Your Environment

If your workspace is behaving unpredictably or you have installed incompatible Python packages:

1. Stop your server via the Hub Control Panel.

2. Log out and log back in.

3. If issues persist, contact an administrator to have your user volume cleared (Note: This will delete your notebooks, so ensure you have a backup).

> **Note**
>
> Always check the **Explorer** first. If the Explorer is down, it is likely a system-wide maintenance issue.

8 License

This project is a fork of https://github.com/opendatacube/cube-in-a-box, and consequently inherits its **MIT License**.

Figure 8.1: License: MIT

You are free to use, modify, and distribute this software under the terms of the MIT License. For more details, see the full license text: MIT[1].

[1] https://opensource.org/license/mit